Ali Mezher

Curriculum Vitae

Department of Mathematics
The University of Texas at Austin
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Research interests

Mathematical physics and analysis of nonlinear PDEs; scaling and mean-field limits of quantum dynamical systems; spectral and dynamical problems in quantum field theory; coupled particle–radiation dynamics, nonlinear resonance cascades, and radiative Bose–Einstein condensation. Current research interests also include entropy vanishing, purification of quantum states, and local well-posedness for nonlinear dispersive equations.

Education

2023–present **Ph.D. studies in Mathematics,** *The University of Texas at Austin*
Department of Mathematics, College of Natural Sciences.
Research areas: mathematical physics and analysis of PDEs.
Advisor: Thomas Chen.

2020–2023 **M.Sc. in Applied Mathematics,** *Kyoto University*
Research Institute for Mathematical Sciences (RIMS).
Specialization: mathematical analysis of nonlinear dispersive PDEs.
Advisor: Kenji Nakanishi.

2015–2018 **M.Sc. in Applied Mathematics,** *University of Kufa*
Faculty of Computer Science and Mathematics.
Research focus: ordinary differential equations in the complex domain.

2005–2009 **B.Sc. in Mathematical Sciences,** *University of Al-Qadisiyah*
College of Science.

Honors and fellowships

2025–2026 **Graduate School Continuing Excellence Fellowship**
College of Natural Sciences, The University of Texas at Austin.

2020 **Japanese Government (MEXT) Scholarship**
Selected in 2020 for one of three scholarship places offered to Iraq by Japan's Ministry of Education, Culture, Sports, Science and Technology.

Publications and preprints

Earlier publications appeared under the name Ali K. Joohy.

Preprints and submitted manuscripts

– Thomas Chen and **Ali Mezher**. Dynamical BEC formation and vanishing entropy for trapped bosons in non-relativistic QED via nonlinear resonance cascades. 2026. arXiv:2610.17923. *Submitted*.

– Thomas Chen and **Ali Mezher**. Trapped bosons in mean field QED, nonlinear resonance cascades and dynamical BEC formation. 2026. arXiv:2604.11756. *Submitted*.

– **Ali Mezher**. On the modified Zakharov–Kuznetsov equation on the cylinder: A trilinear Bourgain-space estimate and its application to local well-posedness. 2026 revision. arXiv:2407.04850. *Submitted*.

Journal articles

– Ali K. Joohy **Ali Mezher**, Ghassan A. Al-Juaifri, and Mohammed S. Mechee. An Investigation of Solving Third-Order Nonlinear Ordinary Differential Equation in Complex Domain by Generalising Prelle–Singer Method. *International Journal of Differential Equations*, 2020, Article ID 5276024.

– Mohammed S. Mechee, Ghassan A. Al-Juaifri, and Ali K. Joohy (**Ali Mezher**). Modified homotopy perturbation method for solving generalized linear complex differential equations. *Applied Mathematical Sciences*, 11(51), 2527–2540, 2017.

Books and Lecture notes

– Ali K. Joohy (**Ali Mezher**). *Ordinary Differential Equations in Complex Domain with Applications*. Scholars' Press, 2018. ISBN: 978-620-2-31082-6.
Related manuscript: arXiv:1805.03201.

Research presentations

Sept. 4, 2026 **Dynamical Relaxation in Non-Relativistic QED: Resonances and Metastable States**
J. Analysis Seminar, Department of Mathematics, The University of Texas at Austin.

Summer schools and academic training

June 2024 **Particle Interactive Systems: Analysis and Computational Methods**
Summer Graduate School, Simons Laufer Mathematical Sciences Institute (SLMath), Berkeley, California. June 17–28, 2024. Participant.

Teaching experience

2023–present **Teaching Assistant,** *The University of Texas at Austin*
Department of Mathematics. Courses and responsibilities include:

Fall 2026 Differential Equations and Linear Algebra

Spring 2026 Discrete Mathematics; Linear Algebra and Matrix Theory.

Spring/Fall 2025 Scientific Computations in Numerical Analysis.

Fall 2024; Summer 2025 Differential Equations with Linear Algebra.

Fall 2023; Spring 2024 Probability I.

Multivariable Calculus.

2021–2023 **Teaching Assistant,** *Kyoto University*
Advanced Calculus (Analysis I) with Karel Svadlenka (2022–2023);
Linear Algebra with Benoit Collins (2021–2022).

Teaching experience (continued)

2019; 2024–2025 **Independent Online Instructor**
Private mathematics courses, including undergraduate and graduate instruction:

2025 Real Analysis (graduate course).

2025 Predictive Modeling for Engineering (graduate course).

2024 Introduction to Nonlinear Dispersive Equations.

2019 Introduction to Real Analysis.

2011–2020 **Mathematics Teacher,** *Ministry of Education, Iraq*
Selected to teach mathematics to final-year secondary-school students. Selected five times to supervise national final examinations.

Educational leadership and outreach

2016–2020 **Founder and Instructor,** *Private Teaching Center, Thi-Qar, Iraq*
Established a center offering mathematics and English instruction; taught part-time.

- Taught Calculus I and II and Linear Algebra to engineering and science students.
- Taught English at levels A1–C1, covering reading, writing, listening, and speaking, using the *New Headway English Course* series.

Est. 2014 **Educational Content Creator,** *Ali Joohy YouTube Channel*
Developed freely accessible mathematics and English courses. The Arabic mathematics portfolio includes real analysis, ordinary differential equations and systems, Calculus I–III, probability and statistics, abstract algebra, classical mechanics, discrete mathematics, and mathematical logic.

Volunteer Teaching
Provided free supplementary mathematics lectures for secondary-school students and syllabus-based English lessons for final-year secondary-school students.

Additional professional experience

2009–2011 **English–Arabic Interpreter and Advisor,** *Global Linguistic Solutions*
Imam Ali (Tallil) Air Base, Thi-Qar, Iraq.

- Facilitated communication between U.S. Army platoon leadership and local residents during field missions.
- Coordinated and interpreted formal and informal meetings between U.S. and Iraqi military personnel.
- Interpreted training delivered by U.S. specialists to Iraqi police personnel and translated accompanying lecture notes.
- Translated invitations, announcements, and other correspondence between English and Arabic.

Technical skills and languages

Computing C, Python, Mathematica.

Languages Arabic, English, Japanese (basic).